# Walmart Hurricane Project

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# The Mess: Issues in the Raw Data

## SKU Codes

**4**

**codes for one product**

Strawberry Pop-Tarts appeared as PT-12, SKU#459812, POPTART-STRAW, and PT_STRAWBERRY.

## Store Records

**14 → 12**

**records for 12 real stores**

FL-239 and FL-287 each appeared twice under mismatched IDs (FL239, FL_287).

## Timestamps

**884**

**transactions with no date**

Out of 25,532. Kept, but left out of daily analyses instead of guessing dates.

**Retained 884 transactions with missing timestamps rather than inventing dates; excluded them only from time-based analyses.**

# The Pipeline

## Raw Data

**25,562**

Transactions

30

Exact duplicates removed

## Clean & Standardize

Deduplication
Dates & IDs
SKU mapping

37 → 10

Category labels

## Analysis Ready

**25,532**

Transactions

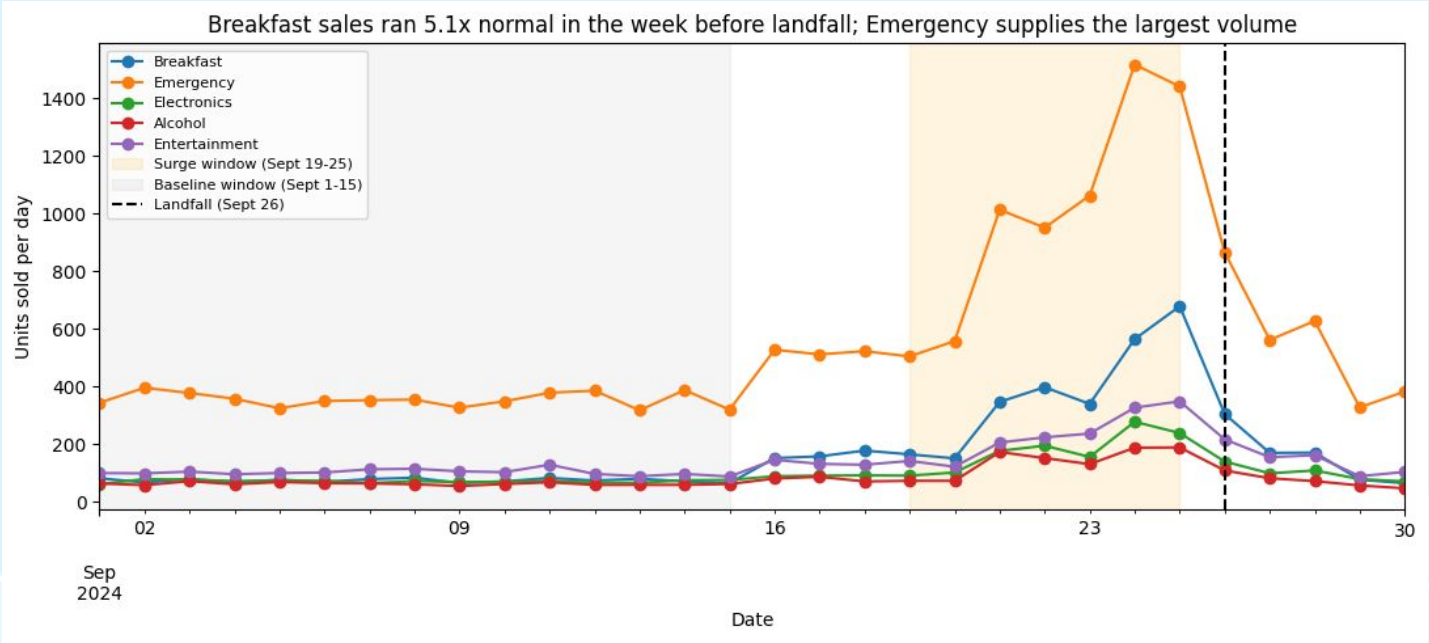
45 → 20

SKUs → Canonical products

**Retained 884 transactions with missing timestamps rather than inventing dates; excluded them only from time-based analyses.**

# Q1: Breakfast Demand Surged 5.14x

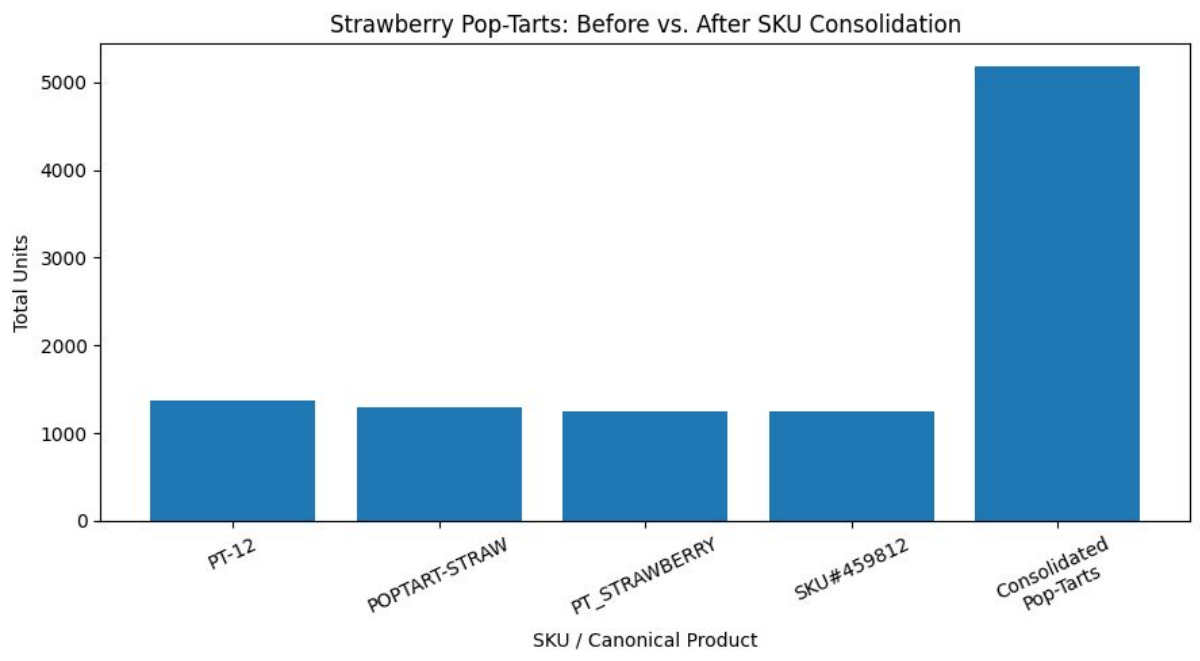
## Breakfast Foods saw the largest pre-hurricane demand increase.




**Pre-storm purchasing increased sharply, making Breakfast Foods a priority for inventory planning.**

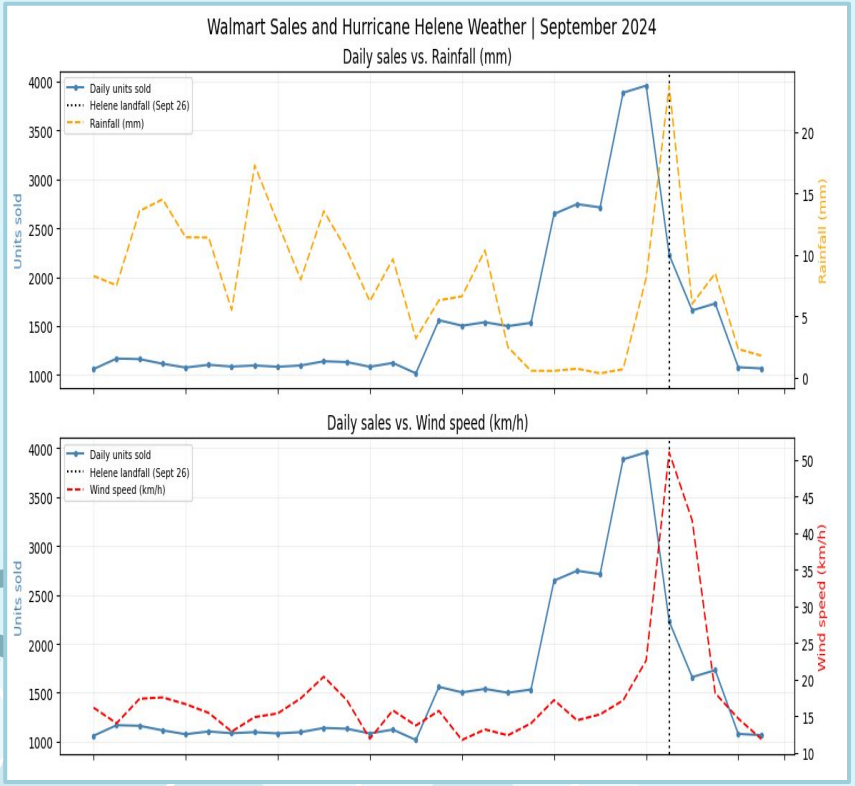
# Q2: SKU Consolidation reveals demand

## Four different SKUS = one product




Strawberry Pop-Tarts totaled 5,178 units across 4 identifiers, not just 1,377 under the largest individual SKU.

# Q3: Weather-Sales Correlation Analysis




## Key Findings

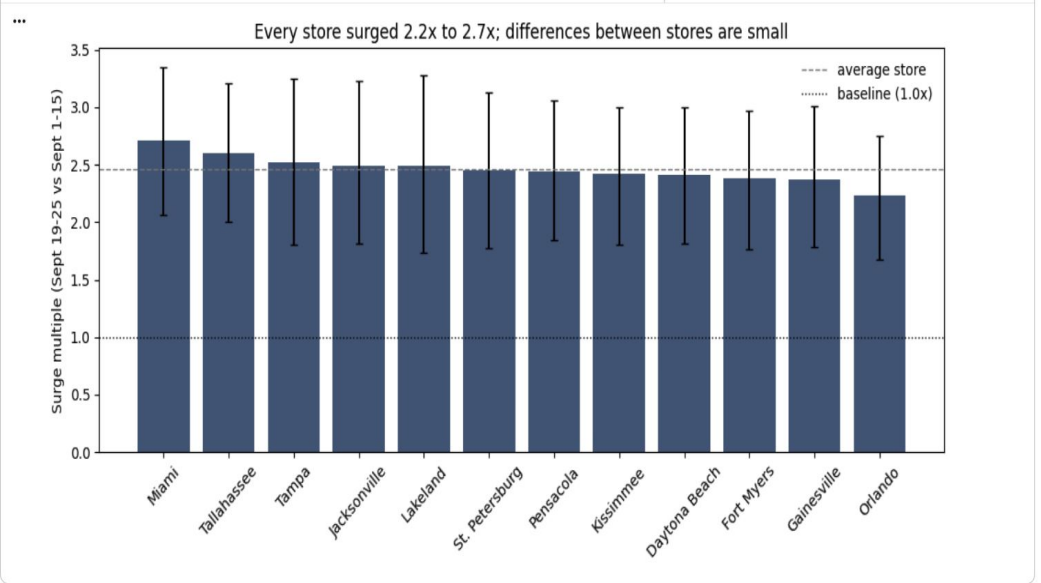
Pre-Storm Sales Surge: Daily sales increased from approximately 1,500 units to nearly 4,000 units before Hurricane Helene's landfall on September 26, suggesting possible emergency stockpiling.

Rainfall Had the Strongest Association: Previous-day rainfall showed the strongest negative correlation with sales (r = −0.550), compared with same-day rainfall (r = −0.298). This suggests a potential delayed relationship between rainfall and consumer purchasing behavior.

Weaker Relationships for Wind and Pressure: Same-day wind speed (r = +0.240) and atmospheric pressure (r = −0.282) showed relatively weak correlations with sales.

# Q4: No geographic pattern in store surges

Distance from landfall did not predict surge size or timing



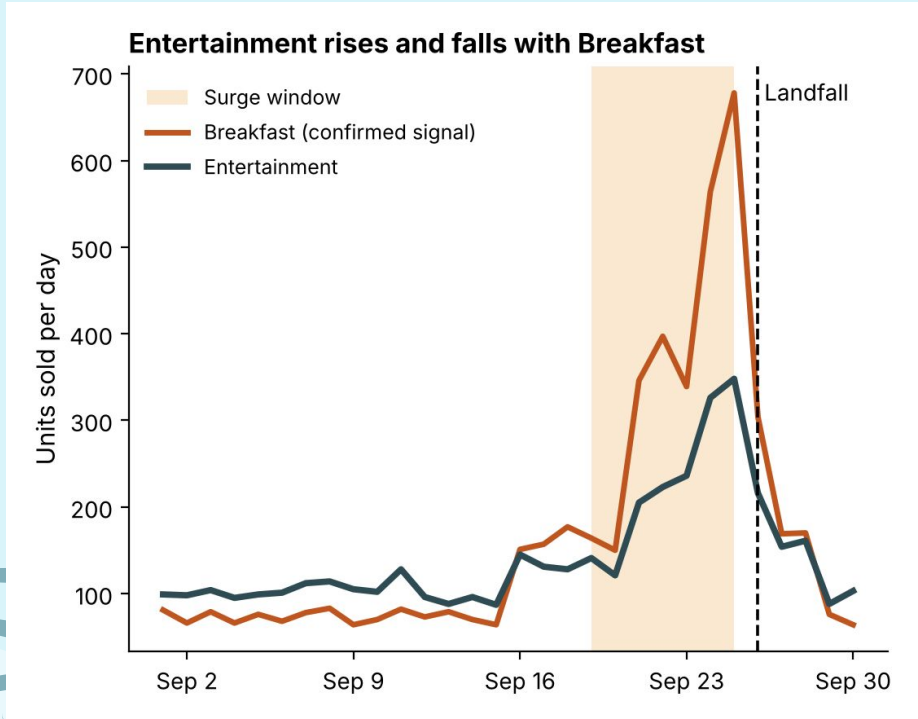

All stores surged alike, no clear outlier. Landfall point at Perry, FL is our assumption.

Baseline window: Sept 1-15
Surge window: Sept 19-25

Plan statewide, not store by store

Error bars: approximate 95% range from resampling each store's days
Nearest store (Tallahassee, 47 mi): 2.60x | Farthest store (Miami, 365 mi): 2.71x | Both peaked Sept 24–25
Distance vs Surge: (r=+0.27, n=12)

# Q5: Board Games are a Real Signal




## Key Findings

Not one SKU: Monopoly rose 2.60x and playing cards 1.99x, each on its own.

Not one store: all 12 stores rose, and the top store was only 10.1% of surge-week units.

Matches a real signal: Entertainment tracked Breakfast day by day (r = 0.98 across 30 days).

Recommendation: pilot a modest board-game increase at selected stores before the next storm. The SQLite database shows the same surge (about 2.5x, peaking Sept 24–25), but its volumes don't match the transactions (0 of 338 store-days), so we're not yet confident enough to stock at scale.

# Recommendation: Stock by unit, not percentages

- Pop-Tarts first (5.1x normal), then emergency supplies, which add more units per day (+650 vs. +300)
- Plan for 4–5x normal, not exactly 5.14x (range 4.3x–7.2x depending on the window)
- Ship to all 12 stores about a week before landfall, from Jacksonville (surge capacity 1,000 shipments/day)
- Trigger on wind and pressure forecasts, since sales led the weather by about 2 days
- Board games: small top-up, then review



# Reflection: What broke, what mattered

## The Challenge

While integrating Walmart sales data with Open-Meteo weather data, we discovered that store FL-619 (Tallahassee, Florida) was missing latitude and longitude coordinates.

Without these coordinates, the store could not be matched to historical weather observations, potentially reducing our analytical coverage.

## Our Solution

Identified the missing coordinates through data validation checks.

Used Tallahassee's approximate city-center coordinates (30.4383, -84.2807) as a documented geographic proxy.

Successfully retrieved weather data for all 12 stores and verified that the merge preserved all 25,532 sales transactions.

## What we learned

Small data-quality issues can have downstream effects on analysis and conclusions.

Documenting assumptions and validating joins are essential for maintaining data integrity.

Preserving records is important, but transparency about approximations is equally necessary.